

ASSIGNMENT - 5

(1) Let Ω_{FS} denote the Fubini-Study form on P^1 .

(a) Find the volume of the symplectic manifold $(\mathbb{C}P^1, \Omega_{FS})$ by evaluating the integral

$$\int_{\mathbb{R}^{2n}} \frac{dx \wedge dy}{(1 + x^2 + y^2)^2}.$$

(b) Note that $\mathbb{C}P^1$ is diffeomorphic with the standard 2 sphere $\mathbb{S}^2$ in $\mathbb{R}^3$ (that is the unit sphere in $\mathbb{R}^3$). Compare the volume of $(\mathbb{C}P^1, \Omega_{FS})$ with the volume of $\mathbb{S}^2$ obtained by integrating the standard volume form of $\mathbb{S}^2$.

You may follow the hint given in Homework 12 (7).

Remark: The standard volume form on $\mathbb{S}^2$ can be expressed in the cylindrical polar coordinate (θ, h) as $\omega_{std} = d\theta \wedge dh$. We have an interesting observation: The area of the sphere $\mathbb{S}^2$ intercepted between $h = h_1$ and $h = h_2$ is the same as the area of the cylinder H intercepted between $h = h_1$ and $h = h_2$, where

$$H = \{(x, y, z) \in \mathbb{R}^3 | x^2 + y^2 = 1, |z| \leq 1\}.$$

(2) Consider $M = \mathbb{R}^6$ with the symplectic form $\omega = \sum_{j=1}^3 dx_j \wedge dy_j$. Let the additive group $G = \mathbb{R}^3$ act on $M = \mathbb{R}^6$ by translation in x coordinates: For $a \in \mathbb{R}^3$, $\psi_a(x, y) = (x+a, y)$ for all $x, y \in \mathbb{R}^3$. Note that we can identify $\mathfrak{g}$ and $\mathfrak{g}^*$ with $\mathbb{R}^3$.

(a) Let $X = a = (a_1, a_2, a_3) \in \mathfrak{g}$. Show that the left invariant vector field $\tilde{X}$ generated by X is given by $\tilde{X}(x) = a$ for all $x \in G = \mathbb{R}^3$.

(b) What is the exponential map for the Lie group G ?

(c) Show that if $X = a = (a_1, a_2, a_3)$, then

$$X^\dagger = a_1 \frac{\partial}{\partial x_1} + a_2 \frac{\partial}{\partial x_2} + a_3 \frac{\partial}{\partial x_3}.$$

(d) Show that the (co)moment map $\mu^* : \mathbb{R}^6 \rightarrow \mathbb{R}^3$ is given by $\mu^*(x, y) = y$, for $x, y \in \mathbb{R}^3$.

(e) Prove that $\mu^a(x, y) = \langle \mu^*(x, y), a \rangle = y \cdot a$. In the expression $\langle \mu^*(x, y), a \rangle$, a is interpreted as a co-vector and an element of $\mathfrak{g}^*$.

(Classically y is called the momentum vector corresponding to the position vector x and these vectors are denoted with an overhead arrow. μ^* is called the *linear momentum*.)

- (3)** The lie algebra of $G = SO(3)$ is the space of all 3×3 skew-symmetric matrices $\mathfrak{o}(n)$. This can be identified with $\mathbb{R}^3$ by

$$A = \begin{pmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{pmatrix}$$

- (a)** If $a, b \in \mathbb{R}^3$ correspond to the skew-symmetric matrices A and B under the above identification then show that $[A, B] = AB - BA$ corresponds to the cross-product $a \times b$ of the vector a and b .
- (b)** Show that the adjoint and the co-adjoint actions of G on $\mathfrak{g}$ and $\mathfrak{g}^*$ respectively are the usual $SO(3)$ action on $\mathbb{R}^3$ by rotation. Show that the orbits are symplectic manifolds.
- (c)** Show that the $SO(3)$ action on $\mathbb{R}^3$ by rotation lifts to the action $\bar{\psi}$ on the cotangent bundle of $\mathbb{R}^3$ as follows:

$$\bar{\psi}_a(x, y) = (a \times x, a \times y)$$

(Use the definition $\bar{\psi}_a \alpha = \psi_{a^{-1}}^* \alpha$ for α in $T^*\mathbb{R}^3$.)

- (d)** Show that the moment map $\mu^* : \mathbb{R}^6 \rightarrow \mathbb{R}^3$ is given by $\mu^*(x, y) = x \times y$.
- (e)** Show that $\mu^a(x, y) = (x \times y) \cdot a$.

The moment map μ^* is called the *angular momentum*.